Cloud Delphi framework under asymmetric uncertainty

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This repository contains the computational implementation and R scripts associated with the article titled *Cloud Delphi framework under asymmetric uncertainty*, which formalizes a generalization of the Cloud model for executing Delphi rounds. The computational procedure automates the processing of expert responses by incorporating the phenomenon of cognitive anisotropy—using split entropies and confidence-based hybrid hyper-entropy operators—to robustly synthesize collective consensus under conditions of asymmetric uncertainty. The pseudocodes appear below. Further information can be requested directly from the author.

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Algorithm 1 AsymmetricCloudRound — synthesis of one Delphi round

Require: Panel data $\{(x_{\text{mín},i}, x_{\text{máx},i}, Ex_i, C_i)\}_{i=1}^{N}$ for round r; previous-round data (if $r > 1$); regularization parameters ϵ, ξ

Ensure: Individual and synthetic metrics $\{Ex_i, En_{L,i}, En_{R,i}, He_i\}_{i=1}^{N}$, $\{Ex_s, En_{Ls}, En_{Rs}, He_s\}$

for $i = 1$ **to** N **do** ▷ per-expert entropy synthesis
  validate $C_i \in [0,1]$ and $x_{\text{mín},i} \le Ex_i \le x_{\text{máx},i}$
  $\phi(C_i) \leftarrow 2 - C_i$ ▷ confidence-driven dispersion factor
  $En_{L,i} \leftarrow \max\big((Ex_i - x_{\text{mín},i})\,\phi(C_i)/3,\ \xi\big)$
  $En_{R,i} \leftarrow \max\big((x_{\text{máx},i} - Ex_i)\,\phi(C_i)/3,\ \xi\big)$
  $A_i \leftarrow \dfrac{|En_{L,i} - En_{R,i}|}{En_{L,i} + En_{R,i} + \epsilon}$ ▷ internal anisotropy index
  $K_{\text{Dynamic}} \leftarrow \max\big((3 + 3C_i)(1 - 0{,}5\,A_i),\ 2\big)$
  $He_i^{\text{base}} \leftarrow (En_{L,i} + En_{R,i})/(2\,K_{\text{Dynamic}})$
  if $r > 1$ **then** ▷ consensus-phase kinematic penalty
    $\Omega_i \leftarrow \big(|x_{\text{mín},i} - x_{\text{mín},i}^{(r-1)}| + |x_{\text{máx},i} - x_{\text{máx},i}^{(r-1)}| + |Ex_i - Ex_i^{(r-1)}|\big)/18$
    $He_i \leftarrow He_i^{\text{base}} + \Omega_i$
  else
    $He_i \leftarrow He_i^{\text{base}}$
  end if
end for
$\overline{Ex} \leftarrow \frac{1}{N}\sum_i Ex_i$ ▷ group synthesis and expert weighting
for $i = 1$ **to** N **do**
  $W_i \leftarrow C_i \Big/ \Big(\big|\frac{Ex_i - \overline{Ex}}{\overline{Ex}}\big| + \frac{En_{L,i} + En_{R,i}}{2} + He_i + \epsilon\Big)$
end for
normalize $W_i \leftarrow W_i / \sum_j W_j$
$Ex_s \leftarrow \sum_i W_i\,Ex_i$; $He_s \leftarrow \sum_i W_i\,He_i$
$En_{Ls} \leftarrow \big(Ex_s - \min_i(Ex_i - 3\,En_{L,i})\big)/3$
$En_{Rs} \leftarrow \big(\max_i(Ex_i + 3\,En_{R,i}) - Ex_s\big)/3$
return $\{Ex_i, En_{L,i}, En_{R,i}, He_i, W_i\}_{i=1}^{N},\ \{Ex_s, En_{Ls}, En_{Rs}, He_s\}$

Algorithm 2 CloudDelphiEvolution — multi-round convergence loop

Require: Sequence of panel data for rounds $1, \dots, R_{\text{máx}}$; thresholds τ, θ; ϵ
Ensure: Round at which the process halts, and its synthetic profile

for $r = 1$ **to** $R_{\text{máx}}$ **do**
  $\{\dots\} \leftarrow$ AsymmetricCloudRound(round-r data, round-$(r-1)$ data)
  $\overline{En}_s^{(r)} \leftarrow (En_{Ls}^{(r)} + En_{Rs}^{(r)})/2$
  $I_s^{(r)} \leftarrow He_s^{(r)} / \overline{En}_s^{(r)}$ ▷ synthetic-cloud uncertainty index
  if $r > 1$ **then**
    $\Delta En \leftarrow \left|\overline{En}_s^{(r)} - \overline{En}_s^{(r-1)}\right| / \text{máx}\left(\overline{En}_s^{(r-1)}, \epsilon\right)$
    $\Upsilon_E \leftarrow [\Delta En < \tau]$
    $\Upsilon_U \leftarrow \left[\, |I_s^{(r)} - I_s^{(r-1)}| < \theta \,\right]$
    if $\Upsilon_E \vee \Upsilon_U \vee (r \geq R_{\text{máx}})$ **then**
      halt; **return** round r and its synthetic profile $\{Ex_s, En_{Ls}, En_{Rs}, He_s\}$
    end if
  end if
end for